

Payoff Formulas and VCG Uniqueness

Jonathan Levin¹

Economics 285
Market Design

Winter 2009

¹These slides are based on Paul Milgrom's.

The Envelope Theorem

- Envelope theorems are about properties of the value function:

$$V(t) \triangleq \max_{x \in X} f(x, t)$$

- Standard envelope theorem: if everything is differentiable...

$$\begin{aligned} V(t) &= f(x(t), t) \\ V'(t) &= f_1(x(t), t)x'(t) + f_2(x(t), t) \\ &= f_2(x(t), t). \end{aligned}$$

- For mechanism design, X may be discrete, or arbitrary in fact.

Milgrom-Segal Envelope Theorems

- Let X be the choice set, $[0, 1]$ be the parameter set.
- Define $V(t) = \max_{x \in X} f(x, t)$, and $X^*(t) = \arg \max_{x \in X} f(x, t)$.

Theorem

Suppose $x \in X^(t)$ and $f_2(x, t)$ exists. If for $t \in (0, 1)$, $V'(t)$ exists, then $V'(t) = f_2(x, t)$.*

Integral Envelope Theorem

- Same environment, X choice set, $[0, 1]$ parameter set.

Theorem

Suppose for all t , $X^(t) \neq \emptyset$, for all (x, t) , $f_2(x, t)$ exists, and that $V(t)$ is absolutely continuous. Then for any selection $x(t)$ from $X^*(t)$,*

$$V(t) = V(0) + \int_0^t f_2(x(s), s) ds.$$

Necessity of the VCG Mechanism

- We will now show that VCG mechanisms are the *only* mechanisms that are both efficient and strategy-proof.
- This result is due to Holmstrom (1979) and for the differentiable case, Green-Laffont (1979).
- We will use the Milgrom-Segal envelope theorem.

- Use broader notion of strategy-proofness (can also proceed by applying the “revelation principle”).
- Let $(N, \{\Sigma_i\}_{i \in N}, \omega)$ be a mechanism.
- Fixing strategy profile σ_{-i} played by $-i$,

$$\begin{aligned} V_i(t_i, \sigma_{-i}) &= \max_{\sigma_i \in \Sigma_i} u_i(\omega(\sigma_i, \sigma_{-i}), t_i) \\ &= V_i(0, \sigma_{-i}) + \int_0^t u_2(\omega(\sigma_i^*(s), \sigma_{-i}), s) ds. \end{aligned}$$

Holmstrom's Lemma

- Suppose $\omega = (z, p)$ specifies decision $z \in Z$ and cash payments p .
- Suppose payoffs are quasi-linear:

$$u_i(z, t_i) - p_i = v_i(t_i) \cdot z - p_i$$

Lemma (Holmstrom)

If payoffs are quasi-linear and agent i has a maximizing choice $\sigma_i^(\cdot)$ independent of others' choices, then*

$$\begin{aligned} V_i(t_i, \sigma_{-i}) &\triangleq v_i(t_i) \cdot z(\sigma_i^*(t_i), \sigma_{-i}) - p_i(\sigma_i^*(t_i), \sigma_{-i}) \\ &= V_i(0, \sigma_{-i}) + \int_0^{t_i} v'_i(s) \cdot z(\sigma_i^*(s), \sigma_{-i}) ds. \end{aligned}$$

Proof of the Holmstrom Lemma

- Agent i 's strategy solves:

$$\max_{\sigma_i \in \Sigma_i} v_i(t_i) \cdot z(\sigma_i, \sigma_{-i}) - p_i(\sigma_i, \sigma_{-i})$$

- The partial derivative of the objective wrt t_i is

$$v_i'(t_i) \cdot z(\sigma_i, \sigma_{-i}).$$

- So by the envelope theorem:

$$V_i(t_i, \sigma_{-i}) = V_i(0, \sigma_{-i}) + \int_0^{t_i} v_i'(s) \cdot z(\sigma_i^*(s), \sigma_{-i}) ds$$

Pinning down payments...

- Re-arranging terms in the Holmstrom Lemma

Corollary

Under the assumptions of the Holmstrom Lemma, the payment rule satisfies:

$$p_i(\sigma_i^*(t_i), \sigma_{-i}) \triangleq v_i(t_i) \cdot z(\sigma_i^*(t_i), \sigma_{-i}) - V_i(0, \sigma_{-i}) - \int_0^{t_i} v'_i(s) \cdot z(\sigma_i^*(s), \sigma_{-i}) ds.$$

Green-Laffont-Holmstrom Theorem

Theorem

Suppose that for each player i , the type space is compact and smoothly path-connected and $v_i(\cdot)$ is continuously differentiable. Then any strategy-proof mechanism that implements the efficient outcome entails the same payments as some Vickrey-Clarke-Groves mechanism.

Proof of GLH Theorem

- Let p be the payment rule of some mechanism that implements the efficient decision performance and let V be its value function.
- Let p^* be a VCG payment function with value function V^* and with $h_i(t_{-i})$ chosen so that:

$$V_i^*(0, t_{-i}) \triangleq V_i(0, \sigma_{-i}(t_{-i})).$$

- Now we apply Holmstrom's Lemma twice...

Proof of GLH Theorem, cont.

- First application:

$$\begin{aligned} p_i(\sigma_i^*(t_i), \sigma_{-i}) &= -V_i(0, \sigma_{-i}) + v_i(t_i) \cdot z(\sigma_i^*(t_i), \sigma_{-i}) \\ &\quad - \int_0^{t_i} v_i'(s) \cdot z(\sigma_i^*(s), \sigma_{-i}) ds. \end{aligned}$$

- Second application:

$$\begin{aligned} p_i^*(t_i, t_{-i}) &= -V_i^*(0, t_{-i}) + v_i(t_i) \cdot z(t_i, t_{-i}) \\ &\quad - \int_0^{t_i} v_i'(s) \cdot z(s, t_{-i}) ds. \end{aligned}$$

- By efficiency, $p_i(\sigma_i^*(t_i), \sigma_{-i}(t_{-i})) \triangleq p_i^*(t_i, t_{-i})$.

Why “path connected”?

- Suppose seller with cost 3 wishes to sell to buyer with value either 2 or 4.
- Consider mechanisms in which parties trade only if trade is efficient and no payment is made if there is no trade.
 - The only such VCG mechanism is the pivot mechanism, at which the price is 3 if the buyer reports a value of 4.
 - ... but the mechanism in which there is trade at a price of 3.5 if the buyer reports a value of 4 is also a dominant strategy mechanism.